

**Columbia agar**

|                                  |             |
|----------------------------------|-------------|
| Pancreatic digest of casein      | 10.0 g      |
| Meat peptic digest               | 5.0 g       |
| Heart pancreatic digest          | 3.0 g       |
| Yeast extract                    | 5.0 g       |
| Maize starch                     | 1.0 g       |
| Sodium chloride                  | 5.0 g       |
| Agar, according to gelling power | 10.0-15.0 g |
| Purified water                   | 1000 mL     |

Hydrate the agar, dissolve by heating to boiling with continuous stirring. If necessary, adjust the pH so that after sterilisation it is  $7.3 \pm 0.2$  at 25 °C. Sterilise in an autoclave using a validated cycle. Allow to cool to 45-50 °C; add, where necessary, gentamicin sulfate corresponding to 20 mg of gentamicin base and pour into Petri dishes.



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**2.6.14. BACTERIAL ENDOTOXINS<sup>(4)</sup>**

The test for bacterial endotoxins (BET) is used to detect or quantify endotoxins from gram-negative bacteria using amoebocyte lysate from the horseshoe crab (*Limulus polyphemus* or *Tachypleus tridentatus*). There are 3 techniques for this test: the gel-clot technique, which is based on gel formation; the turbidimetric technique, based on the development of turbidity after cleavage of an endogenous substrate; and the chromogenic technique, based on the development of colour after cleavage of a synthetic peptide-chromogen complex.

The following 6 methods are described in the present chapter:

- Method A. Gel-clot method: limit test
- Method B. Gel-clot method: quantitative test
- Method C. Turbidimetric kinetic method
- Method D. Chromogenic kinetic method
- Method E. Chromogenic end-point method
- Method F. Turbidimetric end-point method

Proceed by any of the 6 methods for the test. In the event of doubt or dispute, the final decision is made based upon method A unless otherwise indicated in the monograph. The test is carried out in a manner that avoids endotoxin contamination.

**1. APPARATUS**

Depyrogenate all glassware and other heat-stable apparatus in a hot-air oven using a validated process. A commonly used minimum time and temperature is 30 min at 250 °C. If employing plastic apparatus, such as microplates and pipette tips for automatic pipettors, use apparatus shown to be free of detectable endotoxin and which does not interfere in the test.

*NOTE: in this chapter, the term 'tube' includes all types of receptacles, for example microplate wells.*

**2. REAGENTS, TEST SOLUTIONS****(1) Amoebocyte lysate**

Amoebocyte lysate is a lyophilised product obtained from amoebocyte lysate from the horseshoe crab (*Limulus*

*polyphemus* or *Tachypleus tridentatus*). This reagent refers only to a product manufactured in accordance with the regulations of the competent authority.

*NOTE: amoebocyte lysate reacts with some  $\beta$ -glucans in addition to endotoxins. Amoebocyte lysate preparations which do not react with glucans are available; they are prepared by removing from amoebocyte lysate the G factor, which reacts with glucans, or by inhibiting the G factor reacting system of amoebocyte lysate. These preparations may be used for endotoxin testing in the presence of glucans.*

**(2) Lysate solution**

Dissolve amoebocyte lysate in water for BET or in a buffer, as recommended by the lysate manufacturer, by gentle stirring. Store the reconstituted lysate, refrigerated or frozen, as indicated by the manufacturer.

**(3) Water for BET (water for bacterial endotoxins test)**

Water for injections R or water produced by other procedures that shows no reaction with the lysate employed at the detection limit of the reagent.

**3. PREPARATION OF THE STANDARD ENDOTOXIN STOCK SOLUTION**

The standard endotoxin stock solution is prepared from an endotoxin reference standard that has been calibrated against the International Standard, for example *endotoxin standard BRP*.

Endotoxin is expressed in International Units (IU). The equivalence in IU of the International Standard is stated by the World Health Organization.

*NOTE: one International Unit (IU) of endotoxin is equal to one Endotoxin Unit (E.U.).*

Follow the specifications in the package leaflet and on the label for preparation and storage of the standard endotoxin stock solution.

**4. PREPARATION OF THE STANDARD ENDOTOXIN SOLUTIONS**

After vigorously mixing the standard endotoxin stock solution, prepare appropriate serial dilutions of this solution using water for BET.

Use the solutions as soon as possible to avoid loss of activity by adsorption.

**5. PREPARATION OF THE TEST SOLUTIONS**

Prepare the test solutions by dissolving or diluting active substances or medicinal products using water for BET. Some substances or preparations may be more appropriately dissolved or diluted in other aqueous solutions. If necessary, adjust the pH of the test solution (or dilution thereof) so that the pH of the mixture of the lysate and test solution falls within the pH range specified by the lysate manufacturer, usually 6.0 to 8.0. The pH may be adjusted by the use of acid, base or a suitable buffer, as recommended by the lysate manufacturer. Acids and bases may be prepared from concentrates or solids with water for BET in containers free of detectable endotoxin. Buffers must be validated to be free of detectable endotoxin and interfering factors.

**6. DETERMINATION OF THE MAXIMUM VALID DILUTION**

The Maximum Valid Dilution (MVD) is the maximum allowable dilution of a sample at which the endotoxin limit can be determined. Determine the MVD using the following formulae:

$$\text{MVD} = \frac{\text{endotoxin limit} \times \text{concentration of test solution}}{\lambda}$$

(4) This chapter has undergone pharmacopoeial harmonisation. See chapter 5.8. *Pharmacopoeial harmonisation*.

**Endotoxin limit:** the endotoxin limit for active substances administered parenterally, defined on the basis of dose, is equal to:

$$\frac{K}{M}$$

- $K$  = threshold pyrogenic dose of endotoxin per kilogram of body mass,  
 $M$  = maximum recommended bolus dose of product per kilogram of body mass.

When the product is to be injected at frequent intervals or infused continuously,  $M$  is the maximum total dose administered in a single hour period.

The endotoxin limit for active substances administered parenterally is specified in units such as IU/mL, IU/mg, IU/Unit of biological activity, etc., in monographs.

**Concentration of test solution:**

- mg/mL if the endotoxin limit is specified by mass (IU/mg),
- Units/mL if the endotoxin limit is specified by unit of biological activity (IU/Unit),
- mL/mL if the endotoxin limit is specified by volume (IU/mL).

- $\lambda$  = the labelled lysate sensitivity in the gel-clot technique (IU/mL) or the lowest concentration used in the standard curve of the turbidimetric or chromogenic techniques.

## 7. GEL-CLOT TECHNIQUE (METHODS A AND B)

The gel-clot technique allows detection or quantification of endotoxins and is based on clotting of the lysate in the presence of endotoxins. The minimum concentration of endotoxins required to cause the lysate to clot under standard conditions is the labelled lysate sensitivity. To ensure both the precision and validity of the test, confirm the labelled lysate sensitivity and perform the test for interfering factors as described under 1. Preparatory testing.

### 1. PREPARATORY TESTING

#### (i) Confirmation of the labelled lysate sensitivity

Confirm in 4 replicates the labelled sensitivity  $\lambda$ , expressed in IU/mL, of the lysate solution prior to use in the test. Confirmation of the lysate sensitivity is carried out when a new lot of lysate is used or when there is any change in the test conditions which may affect the outcome of the test.

Prepare standard solutions of at least 4 concentrations equivalent to  $2\lambda$ ,  $\lambda$ ,  $0.5\lambda$  and  $0.25\lambda$  by diluting the standard endotoxin stock solution with water for BET.

Mix a volume of the lysate solution with an equal volume of 1 of the standard solutions (such as 0.1 mL aliquots) in each tube. When single test vials or ampoules containing

lyophilised lysate are employed, add solutions of standards directly to the vial or ampoule. Incubate the reaction mixture for a constant period according to the recommendations of the lysate manufacturer (usually at  $37 \pm 1^\circ\text{C}$  for  $60 \pm 2$  min), avoiding vibration. Test the integrity of the gel: for tubes, take each tube in turn directly from the incubator and invert it through approximately  $180^\circ$  in one smooth motion. If a firm gel has formed that remains in place upon inversion, record the result as positive. A result is negative if an intact gel is not formed.

The test is considered valid when the lowest concentration of the standard solutions shows a negative result in all replicate tests.

The end-point is the lowest concentration in the series of decreasing concentrations of standard endotoxin that clots the lysate. Determine the geometric mean end-point concentration by calculating the mean of the logarithms of the end-point concentrations of the 4 dilution series, take the antilogarithm of this value, as indicated by the following expression:

$$\text{Geometric mean end-point concentration} = \text{antilog} \frac{\sum e}{f}$$

$\sum e$  = sum of the  $\log_{10}$  end-point concentrations of the dilution series used,

$f$  = number of replicates.

The geometric mean end-point concentration is the measured sensitivity of the lysate solution (IU/mL). If this is not less than  $0.5\lambda$  and not more than  $2\lambda$ , the labelled sensitivity is confirmed and is used in the tests performed with this lysate.

#### (ii) Test for interfering factors

Prepare solutions A, B, C and D as shown in Table 2.6.14.-1, and use the test solutions at a dilution less than the MVD, not containing any detectable endotoxins, operating as described under 1. Preparatory testing, (i) Confirmation of the labelled lysate sensitivity.

The geometric mean end-point concentrations of solutions B and C are determined using the expression described in 1. Preparatory testing, (i) Confirmation of the labelled lysate sensitivity.

The test for interfering factors must be repeated when any changes are made to the experimental conditions that are likely to influence the result of the test.

The test is considered valid when all replicates of solutions A and D show no reaction and the result of solution C confirms the labelled lysate sensitivity.

If the sensitivity of the lysate determined with solution B is not less than  $0.5\lambda$  and not greater than  $2\lambda$ , the test solution does not contain interfering factors under the experimental conditions used. Otherwise, the test solution interferes with the test.

Table 2.6.14.-1

| Solution | Endotoxin concentration/Solution to which endotoxin is added | Diluent       | Dilution factor | Endotoxin concentration | Number of replicates |
|----------|--------------------------------------------------------------|---------------|-----------------|-------------------------|----------------------|
| A        | None/Test solution                                           | -             | -               | -                       | 4                    |
| B        | $2\lambda$ /Test solution                                    | Test solution | 1               | $2\lambda$              | 4                    |
|          |                                                              |               | 2               | $1\lambda$              | 4                    |
|          |                                                              |               | 4               | $0.5\lambda$            | 4                    |
|          |                                                              |               | 8               | $0.25\lambda$           | 4                    |
| C        | $2\lambda$ /Water for BET                                    | Water for BET | 1               | $2\lambda$              | 2                    |
|          |                                                              |               | 2               | $1\lambda$              | 2                    |
|          |                                                              |               | 4               | $0.5\lambda$            | 2                    |
|          |                                                              |               | 8               | $0.25\lambda$           | 2                    |
| D        | None/Water for BET                                           | -             | -               | -                       | 2                    |

Solution A = solution of the preparation being examined that is free of detectable endotoxins.

Solution B = test for interference.

Solution C = control of the labelled lysate sensitivity.

Solution D = negative control (water for BET).

If the preparation being examined interferes with the test at a dilution less than the MVD, repeat the test for interfering factors using a greater dilution, not exceeding the MVD. The use of a more sensitive lysate permits a greater dilution of the preparation being examined and this may contribute to the elimination of interference.

Interference may be overcome by suitable validated treatment, such as filtration, neutralisation, dialysis or heat treatment. To establish that the treatment chosen effectively eliminates interference without loss of endotoxins, repeat the test for interfering factors using the preparation being examined to which the standard endotoxin has been added and which has then been submitted to the chosen treatment.

## 2. LIMIT TEST (METHOD A)

### (i) Procedure

Prepare solutions A, B, C and D as shown in Table 2.6.14.-2, and perform the test on these solutions following the procedure described under 1. Preparatory testing, (i) Confirmation of the labelled lysate sensitivity.

Table 2.6.14.-2

| Solution | Endotoxin concentration/Solution to which endotoxin is added | Number of replicates |
|----------|--------------------------------------------------------------|----------------------|
| A        | None/Diluted test solution                                   | 2                    |
| B        | 2λ/Diluted test solution                                     | 2                    |
| C        | 2λ/Water for BET                                             | 2                    |
| D        | None/Water for BET                                           | 2                    |

Prepare solution A and solution B (positive product control) using a dilution not greater than the MVD and treatments as described in 1. Preparatory testing, (ii) Test for interfering factors. Solutions B and C (positive controls) contain the standard endotoxin at a concentration corresponding to twice the labelled lysate sensitivity. Solution D (negative control) consists of water for BET.

### (ii) Interpretation

The test is considered valid when both replicates of solution B and C are positive and those of solution D are negative.

When a negative result is found for both replicates of solution A, the preparation being examined complies with the test.

When a positive result is found for both replicates of solution A, the preparation being examined does not comply with the test.

When a positive result is found for one replicate of solution A and a negative result is found for the other, repeat the test.

In the repeat test, the preparation being examined complies with the test if a negative result is found for both replicates of solution A. The preparation does not comply with the test if a positive result is found for one or both replicates of solution A.

However, if the preparation does not comply with the test at a dilution less than the MVD, the test may be repeated using a greater dilution, not exceeding the MVD.

## 3. QUANTITATIVE TEST (METHOD B)

### (i) Procedure

The test quantifies bacterial endotoxins in the test solution by titration to an end-point. Prepare solutions A, B, C and D as shown in Table 2.6.14.-3, and test these solutions according to the procedure described under 1. Preparatory testing, (i) Confirmation of the labelled lysate sensitivity.

### (ii) Calculation and interpretation

The test is considered valid when the following 3 conditions are met:

- (a) both replicates of solution D (negative control) are negative,
- (b) both replicates of solution B (positive product control) are positive,
- (c) the geometric mean end-point concentration of solution C is in the range of 0.5λ to 2λ.

To determine the endotoxin concentration of solution A, calculate the end-point concentration for each replicate, by multiplying each end-point dilution factor by λ.

The endotoxin concentration in the test solution is the end-point concentration of the replicates. If the test is conducted with a diluted test solution, calculate the concentration of endotoxin in the original solution by multiplying the result by the dilution factor.

If none of the dilutions of the test solution is positive in a valid test, report the endotoxin concentration as less than λ (or, if a diluted sample was tested, report as less than the lowest dilution factor of the sample × λ). If all dilutions are positive, the endotoxin concentration is reported as equal to or greater than the largest dilution factor multiplied by λ (e.g. in Table 2.6.14.-3, the initial dilution factor × 8 × λ).

The preparation being examined meets the requirements of the test if the endotoxin concentration in both replicates is less than that specified in the monograph.

Table 2.6.14.-3

| Solution | Endotoxin concentration/Solution to which endotoxin is added | Diluent       | Dilution factor | Endotoxin concentration | Number of replicates |
|----------|--------------------------------------------------------------|---------------|-----------------|-------------------------|----------------------|
| A        | None/Test solution                                           | Water for BET | 1               | -                       | 2                    |
|          |                                                              |               | 2               | -                       | 2                    |
|          |                                                              |               | 4               | -                       | 2                    |
|          |                                                              |               | 8               | -                       | 2                    |
| B        | 2λ/Test solution                                             |               | 1               | 2λ                      | 2                    |
| C        | 2λ/Water for BET                                             | Water for BET | 1               | 2λ                      | 2                    |
|          |                                                              |               | 2               | 1λ                      | 2                    |
|          |                                                              |               | 4               | 0.5λ                    | 2                    |
|          |                                                              |               | 8               | 0.25λ                   | 2                    |
| D        | None/Water for BET                                           | -             | -               | -                       | 2                    |

Solution A = test solution at the dilution, not exceeding the MVD, with which the test for interfering factors was carried out. Subsequent dilution of the test solution must not exceed the MVD. Use water for BET to make a dilution series of 4 tubes containing the test solution at concentrations of 1, 1/2, 1/4 and 1/8, relative to the dilution used in the test for interfering factors. Other dilutions up to the MVD may be used as appropriate.

Solution B = solution A containing standard endotoxin at a concentration of 2λ (positive product control).

Solution C = a dilution series of 4 tubes of water for BET containing the standard endotoxin at concentrations of 2λ, λ, 0.5λ and 0.25λ.

Solution D = water for BET (negative control).

## 8. PHOTOMETRIC QUANTITATIVE TECHNIQUES (METHODS C, D, E AND F)

### 1. TURBIDIMETRIC TECHNIQUE (METHODS C AND F)

This technique is a photometric test to measure the increase in turbidity. Based on the test principle employed, this technique may be classified as being either the end-point-turbidimetric test or the kinetic-turbidimetric test.

The end-point-turbidimetric test (Method F) is based on the quantitative relationship between the endotoxin concentration and the turbidity (absorbance or transmission) of the reaction mixture at the end of an incubation period.

The kinetic-turbidimetric test (Method C) is a method to measure either the time (onset time) needed for the reaction mixture to reach a predetermined absorbance or transmission, or the rate of turbidity development.

The test is carried out at the incubation temperature recommended by the lysate manufacturer (usually  $37 \pm 1^\circ\text{C}$ ).

### 2. CHROMOGENIC TECHNIQUE (METHODS D AND E)

This technique is used to measure the chromophore released from a suitable chromogenic peptide by the reaction of endotoxins with the lysate. Depending on the test principle employed, this technique may be classified as being either the end-point-chromogenic test or the kinetic-chromogenic test.

The end-point-chromogenic test (Method E) is based on the quantitative relationship between the endotoxin concentration and the quantity of chromophore released at the end of an incubation period.

The kinetic-chromogenic test (Method D) measures either the time (onset time) needed for the reaction mixture to reach a predetermined absorbance, or the rate of colour development.

The test is carried out at the incubation temperature recommended by the lysate manufacturer (usually  $37 \pm 1^\circ\text{C}$ ).

### 3. PREPARATORY TESTING

To assure the precision or validity of the turbidimetric and chromogenic techniques, preparatory tests are conducted to show that the criteria for the standard curve are satisfied and that the test solution does not interfere with the test.

Validation of the test method is required when any changes are made to the experimental conditions that are likely to influence the result of the test.

#### (i) Assurance of criteria for the standard curve

The test must be carried out for each lot of lysate reagent.

Using the standard endotoxin solution, prepare at least 3 endotoxin concentrations within the range indicated by the lysate manufacturer to generate the standard curve. Perform the test using at least 3 replicates of each standard endotoxin solution as recommended by the lysate manufacturer (volume ratios, incubation time, temperature, pH, etc.).

If the desired range is greater than  $2 \log_{10}$  in the kinetic methods, additional standards must be included to bracket each  $\log_{10}$  increase in the range of the standard curve.

The absolute value of the correlation coefficient,  $|r|$ , must be greater than or equal to 0.980, for the range of endotoxin concentrations set up.

#### (ii) Test for interfering factors

Select an endotoxin concentration at or near the middle of the endotoxin standard curve.

Prepare solutions A, B, C and D as shown in Table 2.6.14.-4. Perform the test on at least 2 replicates of these solutions as recommended by the lysate manufacturer (volume of test solution and lysate solution, volume ratio of test solution to lysate solution, incubation time, etc.).

Table 2.6.14.-4

| Solution | Endotoxin concentration                                                   | Solution to which endotoxin is added | Number of replicates               |
|----------|---------------------------------------------------------------------------|--------------------------------------|------------------------------------|
| A        | None                                                                      | Test solution                        | Not less than 2                    |
| B        | Middle concentration of the standard curve                                | Test solution                        | Not less than 2                    |
| C        | At least 3 concentrations (lowest concentration is designated $\lambda$ ) | Water for BET                        | Each concentration not less than 2 |
| D        | None                                                                      | Water for BET                        | Not less than 2                    |

Solution A = test solution, that may be diluted not to exceed the MVD.

Solution B = preparation to be examined at the same dilution as solution A, containing added endotoxin at a concentration equal to or near the middle of the standard curve.

Solution C = standard endotoxin solution at the concentrations used in the validation of the method as described under 3. Preparatory testing, (i) Assurance of criteria for the standard curve (positive controls).

Solution D = water for BET (negative control).

The test is considered valid when the following conditions are met:

- the absolute value of the correlation coefficient of the standard curve generated using solution C is greater than or equal to 0.980;
- the result with solution D does not exceed the limit of the blank value required in the description of the lysate reagent employed, or it is less than the endotoxin detection limit of the lysate reagent employed.

Calculate the mean recovery of the added endotoxin by subtracting the mean endotoxin concentration in the solution (if any) (solution A, Table 2.6.14.-4) from that in the solution containing the added endotoxin (solution B, Table 2.6.14.-4).

The test solution is considered free of interfering factors if under the conditions of the test, the measured concentration of the endotoxin added to the test solution is within 50-200 per cent of the known added endotoxin concentration, after subtraction of any endotoxin detected in the solution without added endotoxin.

When the endotoxin recovery is out of the specified range, the test solution is considered to contain interfering factors. Repeat the test using a greater dilution, not exceeding the MVD. Furthermore, interference of the test solution or diluted test solution not to exceed the MVD may be eliminated by suitable validated treatment, such as filtration, neutralisation, dialysis or heat treatment. To establish that the treatment chosen effectively eliminates interference without loss of endotoxins, repeat the test for interfering factors using the preparation being examined to which the standard endotoxin has been added and which has then been submitted to the chosen treatment.

### 4. TEST

#### (i) Procedure

Follow the procedure described in 3. Preparatory testing,

(ii) Test for interfering factors.

#### (ii) Calculation

Calculate the endotoxin concentration of each replicate of solution A using the standard curve generated by the positive control solution C.

The test is considered valid when the following 3 requirements are met:

- (1) the results obtained with solution C comply with the requirements for validation defined under 3. Preparatory testing, (i) Assurance of criteria for the standard curve,
- (2) the endotoxin recovery, calculated from the endotoxin concentration found in solution B after subtracting the endotoxin concentration found in solution A, is within the range of 50-200 per cent,

(3) the result obtained with solution D (negative control) does not exceed the limit of the blank value required in the description of the lysate employed, or it is less than the endotoxin detection limit of the lysate reagent employed.

### (iii) Interpretation

The preparation being examined complies with the test if the mean endotoxin concentration of the replicates of solution A, after correction for dilution and concentration, is less than the endotoxin limit for the product.

*Guidelines on the test for bacterial endotoxins are given in general chapter 5.1.10.*

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## 2.6.15. PREKALLIKREIN ACTIVATOR

Prekallikrein activator (PKA) activates prekallikrein to kallikrein and may be assayed by its ability to cleave a chromophore from a synthetic peptide substrate so that the rate of cleavage can be measured spectrophotometrically and the concentration of PKA calculated by comparison with a reference preparation calibrated in International Units.

The International Unit is the activity of a stated amount of the International Standard, which consists of freeze-dried prekallikrein activator. The equivalence in International Units of the International Standard is stated by the World Health Organization.

### REAGENTS

*Prekallikrein activator in albumin BRP* is calibrated in International Units by comparison with the International Standard.

**Buffer A.** Dissolve 6.055 g of *tris(hydroxymethyl)aminomethane R*, 1.17 g of *sodium chloride R*, 50 mg of *hexadimethrine bromide R* and 0.100 g of *sodium azide R* in *water R*. Adjust to pH 8.0 with 2 M *hydrochloric acid R* and dilute to 1000 mL with *water R*.

**Buffer B.** Dissolve 6.055 g of *tris(hydroxymethyl)aminomethane R* and 8.77 g of *sodium chloride R* in *water R*. Adjust to pH 8.0 with 2 M *hydrochloric acid R* and dilute to 1000 mL with *water R*.

### PREPARATION OF PREKALLIKREIN SUBSTRATE

*To avoid coagulation activation, blood or plasma used for the preparation of prekallikrein must come into contact only with plastics or silicone-treated glass surfaces.*

Draw 9 volumes of human blood into 1 volume of anticoagulant solution (ACD, CPD or a 38 g/L solution of *sodium citrate R*) to which 1 mg/mL of *hexadimethrine bromide R* has been added. Centrifuge the mixture at 3600 g for 5 min. Separate the plasma and centrifuge again at 6000 g for 20 min to sediment platelets. Separate the platelet-poor plasma and dialyse against 10 volumes of buffer A for 20 h. Apply the dialysed plasma to a chromatography column containing *agarose-DEAE for ion-exchange chromatography R* that has been equilibrated in buffer A and is equal to twice the volume of the plasma. Elute from the column with buffer A at 20 mL/cm<sup>2</sup>/h. Collect the eluate in fractions and record the absorbance at 280 nm (2.2.25). Pool the fractions containing the 1<sup>st</sup> protein peak so that the volume of the pool is about 1.2 times the volume of the platelet-poor plasma.

Test the substrate pool for absence of kallikrein activity by mixing 1 part with 20 parts of the pre-warmed chromogenic substrate solution to be used in the assay and incubate at 37 °C for 2 min. The substrate is suitable if the increase in absorbance is less than 0.001 per minute. Add to the pooled

solution 7 g/L of *sodium chloride R* and filter through a membrane filter (nominal pore size 0.45 µm). Freeze the filtrate in portions and store at – 25 °C; the substrate may be freeze-dried before storage.

Carry out all procedures from the beginning of the chromatography to freezing in portions during a single working day.

### METHOD

The assay may be carried out using an automated enzyme analyser or a suitable microplate system allowing kinetic measurements, with appropriate software for calculation of results. Standards, samples and prekallikrein substrate may be diluted as necessary using buffer B.

Incubate diluted standards or samples with prekallikrein substrate for 10 min such that the volume of the undiluted sample does not exceed 1/10 of the total volume of the incubation mixture to avoid errors caused by variation in ionic strength and pH in the incubation mixture. Incubate the mixture or a part thereof with at least an equal volume of a solution of a suitable synthetic chromogenic substrate, known to be specific for kallikrein (for example, *N-benzoyl-L-prolyl-L-phenylalanyl-L-arginine 4-nitroanilide acetate R* or *D-prolyl-L-phenylalanyl-L-arginine 4-nitroanilide dihydrochloride R*), dissolved in buffer B. Record the rate of change in absorbance per minute for 2–10 min at the wavelength specific for the substrate used. Prepare a blank for each mixture of sample or standard using buffer B instead of prekallikrein substrate.

Depending on the method used, ΔA/min has to be corrected by subtracting the value obtained for the corresponding blank without the prekallikrein substrate. The results may be calculated using a standard curve, a parallel-line or a slope ratio assay or any other suitable statistical method. Plot a calibration curve using the values thus obtained for the reference preparation and the respective concentrations; use the curve to determine the PKA activity of the preparation to be examined. During method validation, the spiking experiments must show that the sample matrix has no influence on the results. High blank values may impact assay validity and should be appropriately investigated.

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## 2.6.16. TESTS FOR EXTRANEIOUS AGENTS IN VIRAL VACCINES FOR HUMAN USE

### INTRODUCTION

A strategy for testing extraneous agents in viral vaccines must be developed based on a risk assessment following the principles of viral contamination risk detailed in general chapter 5.1.7. *Viral safety*. This strategy includes a full package of suitable tests that are able to detect different families of extraneous agents that may infect the source of virus strains including cell substrates and raw material of animal or plant origin. It also takes into account the capacity of the manufacturing process to remove or inactivate viruses. The list of tests summarised in Table 2.6.16.-1 must be adapted depending on the extraneous agents that have the potential to contaminate the product: for *in vitro* tests, the risk assessment may allow, with the agreement of the competent authority, the use of other permissive cell lines or molecular biology methods depending on the manufacturing process and the incubation temperature for the growth of particular viruses. If *in vivo* tests are more relevant than *in vitro* tests for the detection of some adventitious viruses (e.g. suckling mice for the vesicular